

AI Use Disclosure

DA3408 Assignment 3 (Spark vs. Ray) | Aramati Chiru Thejaswi (DA24B036) | 5 October 2026

Tool used: Claude (Anthropic), as a chat assistant, throughout the assignment.

What the AI helped with

- **Code:** drafted the PySpark pipeline (*spark_clean.py*), translated its logic into Ray Data (*ray_clean.py*), and wrote the data-check and output-comparison scripts.
- **Debugging:** explained error messages and suggested commands to diagnose memory and cluster problems.
- **Tuning:** suggested a map-side join in Ray, a matching broadcast join in Spark, optional de-duplication, and reading the zone CSV so "N/A" stays text. These are described in the report's Performance Tuning Note.
- **Writing:** drafted the report text and charts from my measured numbers.

I am responsible for the work I am submitting.